

Supplementary Material for
*Inference, Not Just Optimisation: Streaming Inversion-Free
Quasi-Newton Estimation on Dependent Data Streams*

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This Supplementary Material contains four figures. Each supports a claim that the manuscript also establishes by other means, so the manuscript is self-contained without them; they are collected here because Information Sciences suggests a maximum of ten figures plus tables for a theoretical article, and the main text uses exactly ten. Section, theorem and equation numbers referred to in the captions are those of the manuscript. Every number in the captions is generated from the saved result files in the public repository, <https://github.com/Kendiukhov/streaming-quasi-newton-dependent-data>.

References

fig7_lrv_rate pending

Figure S1: The rate claim, measured, on the object an interval actually uses. Both axes logarithmic; error bars are ± 1.96 Monte Carlo standard errors; fitted slopes in the legends. **(a)** Relative operator-norm error in estimating the whole sandwich $\Sigma = H^{-1}SH^{-1}$. Ours is $\bar{H}^{-1}\hat{S}^{\text{FT}}\bar{H}^{-1}$ at $b \asymp \log N$; the comparator is overlapping batch means on the *iterate* path at the block length $b_n \asymp T^{3/4}$ that analysis requires (the manuscript gives the form), which uses no gradients and no curvature estimate. Same target, so the comparison is meaningful; but read the levels, not the slopes. The iterate-path curve is flat here, and its published $N^{-1/8}$ rate is too slow for our range to resolve, so its fitted slope says nothing about it. What the panel shows is that over the sample sizes we can reach, one estimator is accurate and the other is not. **(b)** Error in estimating S alone, each estimator at its own optimal block length: the two-scale form at $b \asymp \log N$ against the $-1/2$ of the manuscript(iv), plain batch means at $b \asymp N^{1/3}$ against $-1/3$.

fig3_gapping pending

Figure S2: At matched cost, a deterministic gap decorrelates curvature updates exponentially in m while random thinning only does so linearly (the manuscript). Vertical axis: $n \mathbb{E} \|\bar{H} - H\|_F^2$ divided by its exact independent-data value $(\text{tr } \Sigma)^2 + \|\Sigma\|_F^2$ — a computed constant, not a fitted normalisation. Pale bands are the two predictions; markers are measurements with the optimisation frozen at θ^* , so only estimation quality is compared. $\kappa_H = (1 + \phi^2)/(1 - \phi^2)$ here.

fig4_cost pending

Figure S3: The statistically correct interval is the cheaper one. Wall-clock milliseconds for one pass over $N = 200,000$ observations (vertical axis, log scale) against dimension d (horizontal, log scale), single-threaded, minimum over seven interleaved repetitions. The i.i.d. plug-in score covariance costs $O(d^2)$ per *observation*; the blocked long-run covariance costs $O(d^2)$ per *block*. Slopes are fitted on the four largest d ; at this N they are inflated by the one-off warm-up (see the corresponding equation of the manuscript and the manuscript, which also gives the asymptotic exponents), and the two smallest d are dominated by fixed per-block overhead rather than by either term of the corresponding equation of the manuscript. The message of the figure is the vertical separation, which holds at every d .

fig5_ablations pending

Figure S4: Ablations. (a) Coverage converges to nominal in N for the two-scale variance; the dependence-blind interval converges to 0.749, the value predicted by the manuscript. (b) Coverage is flat in the block length for the two-scale variance and degrades at both ends for plain batch means. (c) The curvature warm-up is required: without it the $1/t$ schedule leaves a slowly decaying error that destroys coverage even though the variance estimate is accurate.